# Slick Trace investigation report — case_0006

Generated 2026-09-14T07:21:50.010554+00:00 · reproducible investigation report

**What this document is not.** This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.

## Finding

**No vessel in this scene can account for the observed slick.** No candidate release reproduced the observation above the absolute fit gate. The system returns no name rather than the least-bad one; a source vessel may have left the scene before acquisition.

## Scene

| Field | Value |
|---|---|
| Incident | case_0006 |
| Acquired (UTC) | 2026-09-10T04:30:00+00:00 |
| Sensor | synthetic SAR-like (VV proxy, single channel) |
| Data status | synthetic |
| Pixel spacing | 433.59 m |
| Label source | generator ground truth (synthetic) — not a human or independently confirmed label |

## Evidence layers

| Layer | Provenance |
|---|---|
| AIS | synthetic (fictional MMSIs; no real broadcast data) |
| Forcing | slicktrace_core.synth_generator constant+spatial field v0.2 |
| Licence | Synthetic data generated by this repository; released under the repository licence. No third-party data terms apply. |

## Traffic filtering

| State | Vessels |
|---|---|
| excluded | 2 |
| candidate | 3 |

Unassessed is reported separately from excluded: missing evidence is not exoneration, and collapsing the two would silently convert one into the other.

## Rank fragility

Leader retained in 0.5125 of 80 tested perturbations. A tested perturbation changed the leader; the smallest is recorded in the JSON report.

Absence of a flip is evidence of stability across the tested set, not a proof of stability: single-axis and two-axis perturbations only.

## File integrity

| File | SHA-256 (first 16) |
|---|---|
| attribution_result.json | 8ef0b0ed199a1e5a |
| coverage_result.json | d1ea8c538c42cbdb |
| forcing.npz | 07595f69999ce4c5 |
| forecast_result.json | 6f3ce20d79caf404 |
| fragility_result.json | 8a896e89c0f1a439 |
| historical_mapping.json | f1a8c30015cf9765 |
| manifest.json | 2d6f27e9ab4d9c73 |
| mask_groundtruth.png | e424c14c491b48c9 |
| naive_baseline_result.json | 16bd5fc0659fb747 |
| next_observation_result.json | 378c277f832846b5 |
| scene.png | 36d4a1e0d7b61922 |

Hashes verify file integrity only. They do not establish that an observation is true, that a label is correct, or that this analysis is legally sufficient.

## Uncertainty

• Rankings express physical consistency under tested assumptions. They are not proof of responsibility.

• Ensemble and perturbation-grid win frequencies are evidence scores, not calibrated probabilities. No calibration against held-out real outcomes exists.

• Where no candidate clears the absolute fit gate the system reports unknown source rather than naming the least-bad option.

• Synthetic ranking accuracy is not a measurement of real-world attribution accuracy.

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